

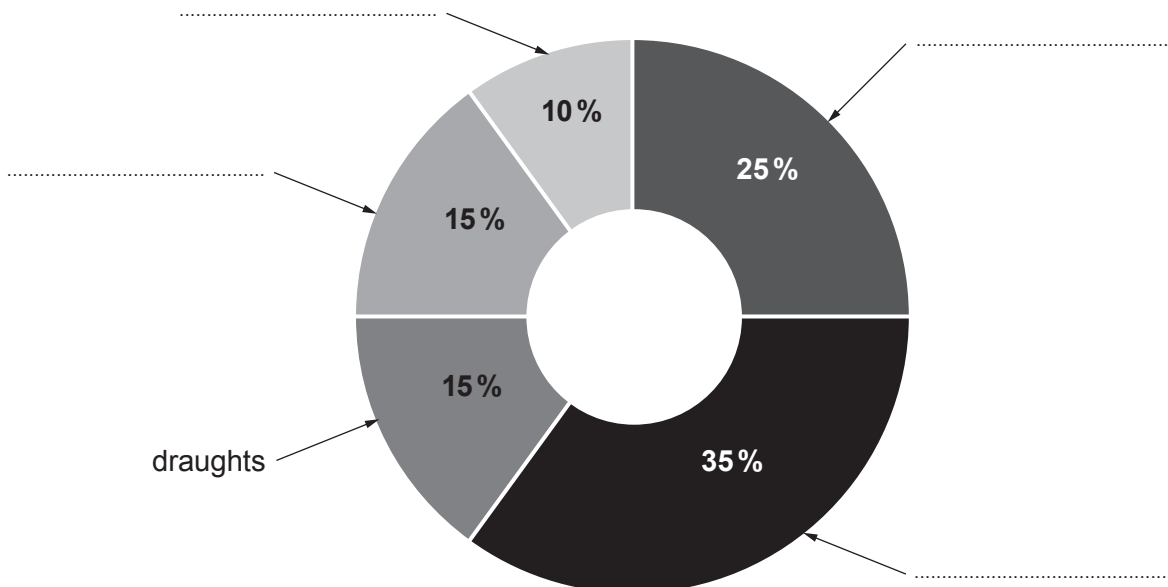
Answer all questions.

1. The Welsh Government Warm Homes Scheme, called Nest, aims to make Welsh homes warmer and more energy efficient places to live. Nest is accessible to all homeowners in Wales and provides advice on saving energy.

- (a) Draughts, floors, windows, walls and the roof are the five ways energy is lost from a heated house. The pie chart shows the percentages of energy loss by each of the five ways.

Use the information below to **complete the labelling** on the pie chart. [3]
One label has been completed already.

- The percentage energy loss through the walls is greatest.
- Windows and draughts have equal percentage energy losses.
- Windows and floor percentage losses add up to equal the percentage loss through the roof.



- (b) To reduce energy loss through the roof the Nest Scheme suggests the installation of fibre-glass insulation in the loft.

- (i) Explain how fibre-glass reduces energy loss in the loft by convection. [2]

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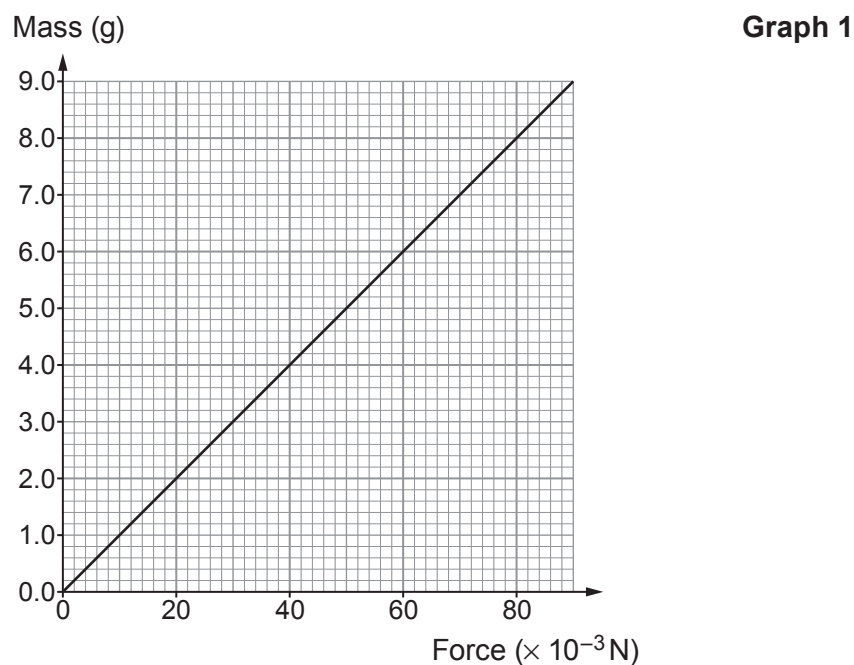
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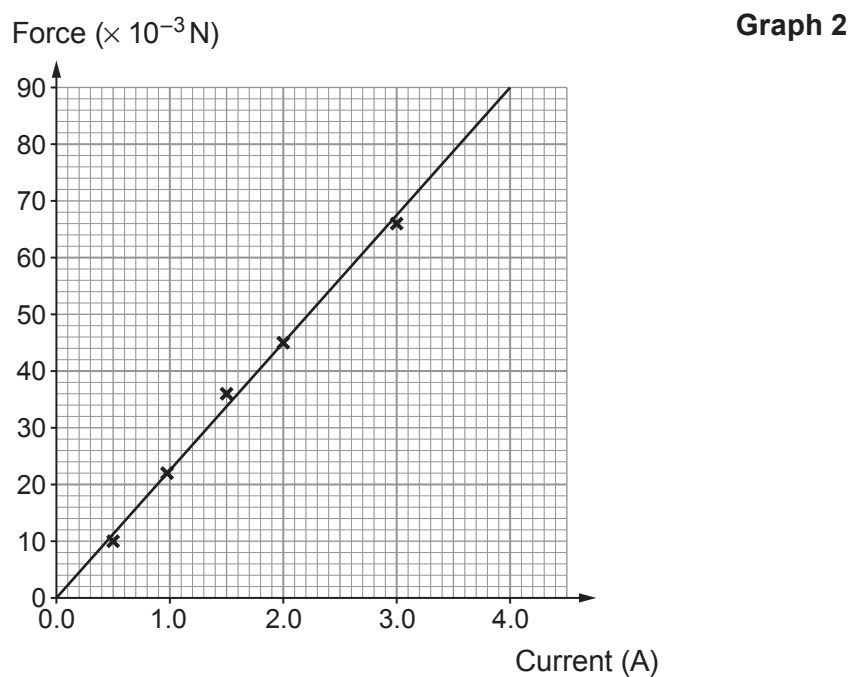
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- (b) When **no** current flows through the wire the digital balance is adjusted to read zero. The mass reading displayed on the digital balance is converted to a force, in newtons, using **Graph 1**.



The current through the wire is varied and the readings of force calculated. The data is plotted on the grid below and a best fit straight line is drawn.



- (i) During the experiment a reading of **4.0 g** is displayed on the digital balance. Use the graphs opposite to complete the table. [2]

Mass (g)	Force (N)	Current (A)
4.0		

- (ii) Describe the relationship between current and force. [2]

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- (iii) A student correctly states that the line on **Graph 2** must obey the equation of a straight line: $y = mx + c$. Comparing this with $F = BIl$ and, given that force is plotted on the y -axis, identify the **quantities** that represent the gradient and the intercept. [2]

Gradient =

Intercept =

- (iv) The length of wire contained in the magnetic field, B , is 5.0 cm. Use information from **Graph 2** to calculate the magnetic field strength of the magnet used in the experiment. [3]

Magnetic field strength, $B =$ T

